

Chin Zhe Ning

Singapore

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EDUCATION

National University of Singapore

College of Humanities and Sciences · (Aug 2022 - May 2026)

- Major in Mathematics (1st) / Statistics (2nd)
- Minor in Computer Science
- GPA: 4.49/5.00
- AY25/26 Semester 1 Dean's List

PROJECTS

Monte Carlo Methods and their Applications in Finance

Capstone project supervised by Prof. Ren Weiqing. · [REPO](#)

- Applied variance reduction techniques for Monte Carlo in rare-event settings such as deep-OTM option pricing and credit risk default probability estimation under multi-factor normal copula models.
- Analyzed estimator efficiency and variance behaviour under extremely low-probability events, with control variates, importance sampling, and control functionals.

Simulation-based Inference in Adaptive Network Epidemic Models

Project for the course Simulation (ST3247) taken under A. Prof. Alexander Thiery. · [REPO](#)

- Applied Approximate Bayesian Computation to fit adaptive epidemic network models with rewiring dynamics to opaque timeseries data.
- Evaluated automatic summary statistic design and calibration quality across inference settings.
- Normalizing flows (SDE model) was used to learn a posterior directly for amortized inference.
- Rewrote the model simulator in C++ for 75× speed-up over the base Python implementation.

WORK EXPERIENCE

CS1010S Teaching Assistant

School of Computing · (Aug 2023 - Dec 2023)

- Conducted weekly tutorials for the course CS1010S: Programming Methodology I, an introduction to programming in Python.
- Break down hard concepts (e.g. recursion) into simple, intuitive explanations and examples.
- Student feedback: 4.20/5.00

Resident Assistant

Ridge View Residential College · (Aug 2024 - May 2026)

- Maintained a safe and inclusive residential community through student engagement and support.
- Acted with residential life staff to respond to crisis situations and ensure timely resolution and follow-up.

SKILLS

- Programming & Tools: Python, C++, and C.
- Scientific Computing: NumPy, SciPy, Pandas, matplotlib, scikit-learn, sbi, PyTorch and JAX.

Relevant Coursework

- [MA4275](#) · Mathematics for Reinforcement Learning
- [MA4270](#) · Data Modelling and Computation
- [MA4230](#) · Matrix Computation
- [MA4251](#) · Stochastic Processes Optimization
- [DSA4212](#) · Optimization for Large-scale Data-driven Inference
- [ST4253](#) · Applied Time Series Analysis

Coursera

- Bayesian Statistics: Techniques and Models (UC Santa Cruz)
- C++ for C Programmers, Part A and B (UC Santa Cruz)
- Approximation Algorithms and Linear Programming (University of Colorado Boulder)
- Mathematics for Machine Learning (Imperial College London)